SARANKUMAR S

Software Engineer | React Native | MERN Stack | Full-Stack Developer

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PROFESSIONAL SUMMARY

Software Engineer with 3.4 years of experience at SRM Tech building and shipping production web, mobile and backend applications with the MERN stack and React Native. Delivered the SmartMeet meeting-intelligence platform, the five-app HealthMate healthcare ecosystem and a no-code chatbot builder, covering real-time WebSocket features, JWT authentication, AWS S3 and Firebase integrations, and REST API development. Works in Agile teams with a focus on reusable components and application performance.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript
Frontend Development: React.js, Redux Toolkit, RTK Query, Vite, Material UI, Recharts, React Flow, Formik, Yup, HTML5, CSS3
Mobile Development: React Native (Android & iOS), react-native-vision-camera, redux-persist, AsyncStorage, Lottie, i18next, Android Studio, Xcode
Backend Development: Node.js, Express.js, REST APIs, WebSockets, Microservices, node-cron, ffmpeg, Joi
Databases: MongoDB, Mongoose, Redis
Cloud & Integrations: AWS S3, Firebase Auth (OTP), FCM Push Notifications, Google Maps API, Google Drive API
Authentication & Security: JWT, Passport JWT, Token Refresh Interceptors, Role-Based Access Control (RBAC)
Development Tools: Git, CI/CD, Postman, VS Code

WORK EXPERIENCE

Software Engineer | SRM Tech, Chennai, Tamil Nadu *June 2023 - Present*

Full-stack engineer delivering web, mobile and backend products for SmartMeet (AI meeting intelligence), TDIP (smart classrooms), HealthMate (healthcare) and SRM Global (chatbot builder).

SmartMeet - Admin & SuperAdmin Web Portal

React.js, TypeScript, Vite, Material UI, Redux Toolkit, RTK Query, Recharts, WebSockets

- Architected a role-based LayoutProvider that renders four UI shells across 20+ screens, removing duplicated layout code.
- Built the SuperAdmin analytics dashboard with Recharts, combining four parallel RTK Query calls and reactive date-range filters to surface user and meeting KPIs.
- Integrated a WebSocket task client for transcription, meeting-status and action-item updates, keeping every dashboard panel current without polling.
- Implemented silent JWT refresh in the RTK Query layer (401, refresh token, retry; redirect to /login on failure) so sessions stay alive without forced re-login.
- Delivered an org-chart editor with CSV/Excel hierarchy import and dirty-state guards, and an IoT device manager with health filters and live metadata (battery, WiFi RSSI, firmware, last 10 logs).

SmartMeet - User Web Portal

React.js, TypeScript, Vite, Material UI, Redux Toolkit, RTK Query, AWS S3, WebSockets

- Built an instant-meeting launcher that streams speaker-labelled live transcripts over WebSockets as they arrive.
- Developed a recording upload flow with AWS S3 progress indicators and WebSocket task-status events, carrying each file from upload through transcription to review.
- Implemented natural-language Advanced Search with RTK Query, filterable by status, date range, platform and participants, surfacing AI summaries and transcript excerpts.
- Brought scheduling, transcript review and action-item tracking together in a single Material UI dashboard.

SmartMeet - Mobile Application

React Native, TypeScript, Redux Toolkit, RTK Query, FCM, AWS S3, WebSockets, Node.js, MongoDB

- Architected a React Native app for three user roles (User, Admin, SuperAdmin) covering scheduling, instant recording, live transcription, AI summaries and action items.
- Built 15+ reusable components (shimmer loaders, date filter strips, audio playback controls, a Lottie device-loading modal) to speed up screen development.
- Engineered FCM push notifications for meeting and action-item events, with Redux state and WebSocket-driven unread badge counts.
- Replaced keyword-only lookup with natural-language semantic search over historical transcripts, so past decisions can be found by intent.
- Resolved production issues: an Android build regression (react-native-audio-recorder-player v3 to v4), an Admin logout bug in inActiveUserSocket, a Mongoose $or key-overwrite bug, and moment.js timezone errors in reminder cron jobs.

SmartMeet TDIP - Teacher Classroom Recording App

React Native 0.84, TypeScript, Redux Toolkit, RTK Query, redux-persist, react-native-vision-camera, Lottie, WebSockets

- Built the Android and iOS faculty app for SRM University's TDIP platform, with QR-based IoT device pairing (react-native-vision-camera and a dedicated /auth/onAuthDevice endpoint) leading straight into live recording.

- Developed dual-token authentication (NxtAgent login plus SmartMeet device token) persisted with redux-persist and AsyncStorage, so sessions survive restarts and reconnects.
- Engineered a RecordingPersistenceService on react-native-fs that saves in-progress recording state, enabling recovery after unexpected app termination.
- Created a ContainerProvider screen wrapper (safe area, header, global loader, NoInternet modal) used across 20+ screens, and a typed library of 70+ custom SVG icon components.

HealthMate - Multi-Role Healthcare Super-App (5 Mobile Apps: Doctor, Patient, Hospital, Lab, Pharmacy)
React Native, TypeScript, Redux Toolkit, RTK Query, Firebase Auth, AWS S3, FCM, Google Maps API

- Built 50+ shared UI components (inputs, modals, bottom sheets, shimmer loaders, PDF/DOCX/image viewers, map pickers) reused across all five apps, cutting per-feature development time by about 60%.
- Engineered dual authentication with Firebase Phone and Email OTP, exchanging Firebase ID tokens for backend JWTs attached by an RTK Query base-query interceptor, with automatic logout on 401.
- Implemented a 5-step doctor onboarding wizard with an admin verification state machine and a withProfileVerification HOC that redirects incomplete profiles to the right setup step.
- Developed a prescription builder that generates branded PDFs (react-native-html-to-pdf) and uploads them to AWS S3 through a centralised upload client also used for lab reports and profile photos.
- Optimised the appointment module (date-strip calendar, Google Maps Places Autocomplete) with useMemo, useCallback and lazy RTK Query hooks, and added offline handling (network detection with an offline modal, persisted Redux state) plus English and Tamil localisation with i18next.

SmartMeet TIPS Middleware - Recording Automation Backend
Node.js, Express.js, TypeScript, MongoDB, Redis, WebSockets, node-cron, Passport JWT, ffmpeg, Google Drive API

- Built a Node.js/Express microservice that automates the path from Google Meet recording to TIPS upload: a per-minute cron job polls Google Drive, converts MP4 to MP3 with ffmpeg, and moves each session through a 14-status, 4-phase workflow (fetch, voice identification, schedule mapping, TIPS upload).
- Integrated an external AI voice-identification service to map speakers to faculty, then matched each faculty member to their timetable slot through the TIPS schedule API, flagging multi-faculty recordings for manual review.
- Implemented admin override tools for manual faculty mapping, schedule retry, session splitting for long recordings, and bulk retry of on-hold sessions filtered by faculty, program, semester and batch.
- Delivered real-time status updates by persisting each status transition as a MongoDB notification and pushing a WebSocket event to the connected admin, and secured the API with Passport JWT, Google service-account impersonation for Drive access, and Redis caching on GET routes.

SRM Global - Chatbot Flow Builder (No-Code Bot Designer)
React, TypeScript, Vite, MUI, RTK Query, React Flow, Formik/Yup, Node.js, Express.js, MongoDB, Joi, AWS S3

- Developed a visual chatbot builder on a drag-and-drop React Flow canvas, supporting node connections, edge reconnection, inserting nodes mid-edge, minimap and zoom controls.
- Built a component-based node editor with property and add-component panels for text, button, image, link, carousel, redirect and workflow components, with reordering and a main-menu node setting.
- Implemented debounced 1-second auto-save, flush-on-navigate, a localStorage draft fallback and publish validation that blocks isolated nodes, so no work is lost and only fully connected flows go live.
- Created a live in-browser ChatPreview that simulates the bot from any start node and highlights the active node on the canvas, plus a 2-step bot setup wizard with S3 logo upload.
- Designed the Express/MongoDB REST API around a single versioned Flow resource with Joi schema validation, graph-connectivity checks, and node- and component-level delete endpoints.

EDUCATION

Diploma in Computer Engineering | Sri Sairam Polytechnic College, Tamil Nadu

CERTIFICATIONS

Anthropic: Introduction to Agent Skills; Claude 101; AI Fluency: Framework & Foundations
freeCodeCamp: Responsive Web Design; Legacy JavaScript Algorithms and Data Structures; Front-End Development Libraries; Back-End Development and APIs
Other: Linux Unhatched

AWARDS & RECOGNITION

- **Rising Star Award** - SRM Tech Accolades Recognition Programme
- **Quality Gyan Quiz Q3 Winner** - recognised for excellence in quality awareness at SRM Tech